Smart Safety Monitoring System For Long Distance Buses In Sri Lanka

Project ID: **25-26J-511**

Smart Safety Monitoring System For Long Distance Buses In Sri Lanka

Problem

Unsafe Long-Distance Bus Transportation in Sri Lanka

- High number of accidents involving long-distance buses.
- Major causes include:
    - Driver fatigue and drowsiness
    - Route and traffic rule violations
    - Passenger overcrowding
    - Delayed accident detection and emergency response
- Lack of real-time monitoring and automated safety systems in buses.
- Dependence on manual supervision, which is often ineffective.

Impact / Gravity

Serious Consequences on Lives and Society

- Loss of human lives and severe injuries
- Psychological trauma to passengers and families
- Economic losses due to:
    - Medical expenses
    - Vehicle damage
    - Traffic delays
- Reduced public trust in long-distance bus services
- Increased burden on hospitals, police, and emergency services
- Negative impact on national road safety statistics

Smart Safety Monitoring System For Long Distance Buses In Sri Lanka

Affected Persons / Groups / Institutions

Directly Affected:

- Passengers using long-distance buses
- Bus drivers and conductors

Indirectly Affected:

- Bus Fleet owners and transport companies
- Road safety authorities and police
- Emergency response teams
- Government and transport regulatory bodies

Who Will Benefit from the Solution:

- Passengers → safer and more reliable travel
- Drivers → fatigue alerts , real-time warnings before danger situation and guidance
- Authorities → accurate, real-time violation tracking and accident reports
- Society → reduced accidents and improved road safety culture

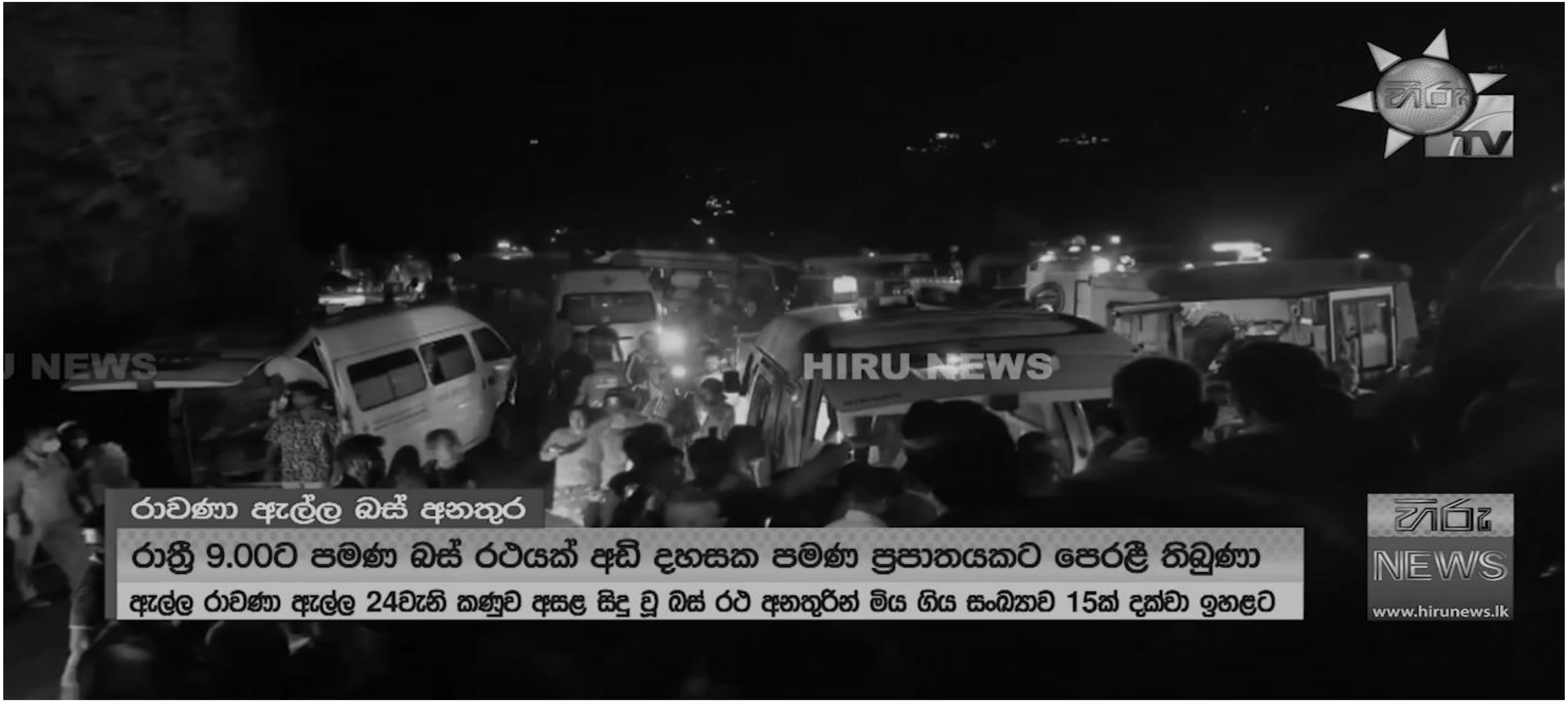
Smart Safety Monitoring System For Long Distance Buses In Sri Lanka

Design Excellence of the Overall Solution

Smart, Practical, and Affordable Safety System

- Real-time monitoring using low-cost sensors and software
- Key features:
  - Driver fatigue detection and alerts
  - Route and rule violation detection
  - Passenger overcrowding monitoring + pre-crash warning system
  - Automatic crash detection
  - Emergency alert system
- Immediate audible alerts to drivers
- Automatic reporting to relevant authorities and fleet owners
- Designed specifically for Sri Lankan road and transport conditions
- Scalable and cost-effective for large-scale deployment

Smart Safety Monitoring System For Long Distance Buses In Sri Lanka



IT22142696 | DIAS D.G.M

Overcrowding Management and Rollover Prediction System for Long-Distance Buses in Sri Lanka

Specialization: Software Engineering

Overcrowding Management and Rollover Prediction System For Long Distance Buses in Sri Lanka

How the solution will address the problem

This system solve three critical safety gaps simultaneously: hazardous overcrowding, unsafe footboard riding, Bus rollovers on curved roads due to high speed and overloading , yet drivers lack real-time visibility into risk.

<u>**My solution is IoT unit that integrates**</u>
Passenger/Footboard Safety by using dual-beam IR sensors to count passengers and detect footboard riding

ML model trained using Physics Engine: Feeds real-time passenger load data into a Python-based physics model. This calculates the **Static Stability Factor (SSF)** against upcoming road curvature (fetched via OSMnx/OpenStreetMap),bus dimensions and center-of-gravity to predict rollover risk before the bus enters a curve.

User requirements addressed by the solution

For Drivers: Real time passenger monitoring , Audible alerts inside the bus to indicate standing on footboard , Predictive lookahead warnings

For Passengers : Live Bus passenger capacity tracking , partially filled bus suggesting on same route, passenger also can check if the driver is driving the bus safely before on board.

For Authorities: Web dashboard provides Violation Analytics by registered buses on the system (Top Offenders),Safety Violation Trends , Real-time Fleet Capacity (Per Bus)

For Conductor : Can check Passenger Capacity Trend , Recent overcrowding and footboard violation records , Can raise maintenance Ticket

Overcrowding Management and Rollover Prediction System For Long Distance Buses in Sri Lanka

Design Excellence / Contribution

- **Hybrid Approach:** Novel integration of Physics-based dynamics (SSF equations) with Geospatial Intelligence (OSMnx mapping).
- **Predictive Accuracy:** Unlike simple accelerometers that react to current force, this system predicts force based on the road ahead.
- **Computational Efficiency:** Optimized to run on edge devices (Raspberry Pi/Jetson) with optional lightweight modes.

Dataset – Since Gathering a Primary Data set is not practical , I developed a Simulation using my physics model to Run Safety Analysis on different routes on Srilanka with variable passenger capacities and speed which generates Rollover Threshold , using that data set I generated a synthetic data set to train my ML model to run Rollover prediction.

I made a machine learning model for the better and faster accuracy instead of directly using physics engine all the time

Task	Problem Type	Model Used	Framework
Predict Rollover Risk Score and Safe Stopping Distance simultaneously.	Multi-Output Regression	Random Forest Regressor	Scikit-Learn

User Feedback on Prototype

"lookahead warnings" provide valuable reaction time compared to traditional sensors, though drivers requested simplified visual alerts (Red/Green) over numerical risk factors.

IT22886286 | DE SILVA U.M.T.K

Route and Road Violation Detection System for Long-Distance Buses in Sri Lanka

Specialization: Software Engineering

Route and Road Violation Detection System For Long Distance Buses In Sri Lanka

How the solution will address the problem

Current public transport system lack a proper mechanism to detect route and road violations throughout a bus's entire journey.

My solution is integrated safety system that detects route deviations and road rule violations in real time, providing immediate audible alerts to the driver and automated notifications to authorities, enabling instant correction and effective enforcement.

User requirements addressed by the solution

For Drivers: Immediate audible alerts inside the bus cabin promote safer driving practices.

For Fleet Operators: Web dashboard provides data-driven insights into driver performance and route adherence, enabling better management.

For Authorities: Automated, evidence-based reporting allows for efficient enforcement of traffic laws, targeting repeat offenders.

Design Excellence / Contribution

Dataset – Gathered Primary Data which suitable for Sri Lankan Roads to achieve better accuracy and use Roboflow to annotate the images and create the dataset.

I made 3 separate models for the better accuracy.

Task	Problem Type	Model Used
Speed-limit detection	Object detection	YOLOv8
Red-light detection	Object detection	YOLOv8
Double-line violation	Image classification	YOLOv8-cls

Made Buzzer device to notify driver when he violate rule.

IT22030030 | SADAN M.D.T

Biometric Security and Driver Monitering System for Long-DistanceBuses in Sri Lanka

Specialization: Software Engineering

Biometric Security and Driver Monitering System for Long-DistanceBuses in Sri Lanka

How the solution will address the problem

Fatigue, unauthorized operation, and prolonged driving without rest are critical safety gaps in the long-distance bus transport system.

My solution provides a high-performance integrated safety layer that ensures only authorized drivers operate the vehicle, monitors real-time fatigue, and prevents overdriving, providing immediate interventions to enhance road safety.

User requirements addressed by the solution

For Drivers: Real-time, non-intrusive monitoring provides instant visual and audible alerts to prevent microsleep or fatigue-related accidents

For Fleet Operators: Secure biometric verification ensures accountability, while a live dashboard displays driver status and alertness, supporting effective fleet management.

For Authorities: Provides a proof log of driver identity, alertness, and work/rest compliance, enabling evidence-based safety audits and enforcement.

Route and Road Violation Detection System For Long Distance Buses In Sri Lanka

Design Excellence / Contribution

The system uses a customized FaceNet model for fast, accurate driver verification and a dual-speed monitoring architecture to track fatigue efficiently. EAR and MAR geometric analysis with 478 MediaPipe landmarks ensures precise drowsiness and yawning detection. Cross-platform data sync via driver_id maintains integrity, while throttled identity checks optimize resource use without compromising safety.

Dataset - The system uses pre-trained FaceNet embeddings for driver identification and MediaPipe facial landmarks for drowsiness detection. This approach ensures accurate recognition and fatigue monitoring further finetune the system, while also handling variations in bus models and camera placement angles for real-world applicability.

Task	Problem Type	Model Used
Driver Identification	Facial Recognition	FaceNet
Drowsiness Detection	Geometric Ratio Analysis	EAR (Eye Aspect Ratio) calculated using MediaPipe facial landmarks
Fatigue/Yawn Detection	Behavioral Analytics	MAR (Mouth Aspect Ratio) calculated using MediaPipe facial landmarks
Performance Optimization	Resource Scheduling	Optimized Processing & Landmark Caching

IT22356154 | GAMLATH G. R. A. A

Emergency Crash Response System for Long-Distance Buses in Sri Lanka

Specialization: Software Engineering

Emergency Crash Response System for Long-Distance Buses in Sri Lanka

Problems and Solutions

Bus accidents in Sri Lanka cause significant losses. The main issues are,

- Delayed emergency response
- Lack of real-time monitoring
- Manual reporting dependency
- Poor accident data

In our system,

- Real-Time Crash Detection
- Automated Alert System
- Severity Classification
- Centralized Dashboard

User requirements addressed by the solution

Transport Authority

- Real-Time Monitoring
- Instant Notification
- Multi-Bus Management
- Accident Analytics

Emergency Services

- Rapid Alert System
- Severity Information
- Crash Verification

Bus Owners

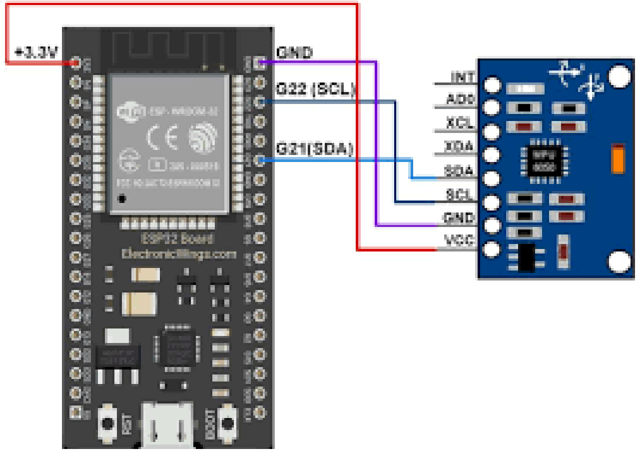
- Driver Behavior Analysis
- Incident Documentation
- Cost-Effective Solution

Emergency Crash Response System for Long-Distance Buses in Sri Lanka

Design Excellence and Key Contributions

- Used Deep Autoencoder Neural Network model.
- Trained to reconstruct normal driving patterns.
- Measures reconstruction error.
- Final result combined,
  - Autoencoder anomaly detection
  - Rule-Based Physics Check

- Hardware and Sensor Components
  - 3-axis gyroscope sensor
  - 3-axis accelerometer sensor
  - ESP32 microcontroller unit (MCU)



Smart Safety Monitoring System For Long Distance Buses In Sri Lanka

Commercialization

- The first integrated safety SaaS platform built for Sri Lanka's transport ecosystem.

- Subscriptions + high-margin data monetization.
    - insurance safety scores, urban planning analytics

- Software-based model easily adapts to new markets across the Country.
    - e.g., trucks, school vans

Thank you !